Data Analytics Capstone Topic Approval Form

Capstone Topic Approval Form

The purpose of this document is to help you clearly explain your capstone topic, project scope, and timeline. Identify each of the following areas so you will have a complete and realistic overview of your project. Your course instructor cannot approve your project topic without this information.

Student Name: Danty Dwayne Cook

Student ID: 007003568

Capstone Project Name: Dataset-Agnostic Condition Monitoring and Anomaly Detection Pipeline for IoT Sensor Data

Project Topic: Condition Monitoring and Anomaly Detection for Predictive Maintenance (Early Warning)

Research Question: Given a standardized, dataset-agnostic preprocessing workflow, does an ensemble of unsupervised anomaly detection models outperform a single unsupervised baseline in detecting abnormal equipment behavior in multivariate IoT sensor time-series data-specifically by reducing false positive while maintaining or improving detection of abnormal performance?

> Hypothesis: Compared with a single unsupervised baseline, an unsupervised ensemble will reduce false positives while maintaining comparable or improved detection of abnormal behavior. False positives will be measured using defensible known-normal operating windows, and detection of abnormal behavior will be measured using available labels and/or controlled anomaly-injection tests.

Context:. This capstone develops a dataset-agnostic condition monitoring and anomaly detection pipeline for multivariate IoT sensor time-series data. The pipeline standardizes ingestion, column/metadata handling, data quality checks, and time/event alignment so that datasets with different schemas can be prepared consistently for modeling.

The primary analytical objective is to compare a single unsupervised anomaly detection baseline against an ensemble of unsupervised anomaly detection models for early-warning identification of abnormal equipment behavior. The comparison will focus on whether the ensemble provides improved robustness and operational usefulness, particularly through reducing false alarms.

Evaluation will be performed using label-based assessment where available. When labels are limited or imperfect, the project will use defensible known-normal operating windows (derived from datatset status fields or documented rules) to estimate false positive counts and to measure alarm burden. Results will be documented through reproducible experiments, metrics, and visualization suitable for review and re-exectuion.

Data: Identify data you will need to collect that are relevant to the situation or question. Publically available industrial IoT/Pump Sensor Dataset by UnknownClass available on Kaggle.com

If an existing dataset will be used, describe the dataset. The dataset consists of multivariate time-series sensor readings from a water pump system for a smaller area away from a bit town. The data was recorded through the use of 52 sensors attached to various components and locations on the Pump system over the course of a year.

The dataset is hosted on Kaggle.com and will be used in accordance with the dataset page terms and Kaggle's Terms of Use. Although the dataset page does not specific a clear open-data license; Therefore, the project will use the dataset in accordance with Kaggle's Terms of Use and not be redistributed beyond what is permitted and all sources will be cited.

Data Gathering: Describe the data-gathering methodology you will use to collect data. Data will be obtained from publically available sources, Kaggle, and imported as CSV files through the use of Python and the Pandas Library. Kafka will also be used to simulate real-time message streaming of sensor observations and model outputs during testing.

Data Analytics Tools and Techniques: Identify the appropriate data-analysis technique you will use to analyze the data. Tools: Data preparation, exploratory analysis, and modeling will be performed using Python and Jupyterlab with the Pandas, Numpy, and Scikit-Learn libraries. Jupyterlab will be used for the EDA step and initial review of the data before handing off to a pure python pipeline.

Techniques: Varies techniques will be utilized through-out the project, such as: data cleaning and quality checks (missing values, duplication checks, invalid ranges, data type conversion), schema standardization, and time/event alignment to support a dataset-agnostic pipeline. Modeling will be done at first through a single unsupervised anomaly detection model to create a baseline then tested against an ensemble of unsupervised models for comparison.

Justification of Tools/Techniques: Explain why the data-analysis technique you chose is an appropriate technique to analyze the data collected. Python, Jupyterlab, and the supporting libraries were selected because they are widely accepted and enable reproducible workflows for cleaning, transforming, and analyzing multivariate time-series sensor data. Schema Standardization, data quality checks, and time/event alignment are used to support dataset-agnostic processing, ensuring the pipline can generalize across IoT datasets with different structures while minimizing false alarms caused by data defects.

Unsupervised Anomly detection models are used because predictive maintenance data often lacks complete, reliable lables. These methods learn patterns of normal operation to detect abnormal behavior and provide early-warning signals that support condition monitoring and predictive maintenance decisions.

Application Type, if applicable (select one):

☐ mobile
☐ web
☒ stand-alone

Programming/Development Language(s), if applicable: Python (Pandas, Numpy, Scikit-learn), JupyterLab, SQL

Operating System(s)/Platform(s), if applicable: Windows, Docker, WSL2, Git/Github, Weights and Biases, Kafka

Database Management System, if applicable: PostgreSQL, Flat Files (Pickle and CSV)

Project Outcomes: List the key anticipated project outcomes and deliverables in fewer than 500 words.
1) A reproducible, dataset-agnostic preprocessing and condition-monitoring pipeline for multivariate IoT sensor time-series data, including schema standardization, data quality validation, and time/event alignment features.
2) A baseline unsupervised anomaly detection model and an unsupervised ensemble implementation;
3) A comparative evaluation of single-model versus ensemble performance with emphasis on false positive reduction and detection of abnormal behavior using known-normal windows and available labels and/or anomaly-injection tests.
4) An optional streaming simulation using Kafka to publish sensor readings and anomaly outputs;
5) Optional storage of processed data and model outputs in PostgreSQL and experiment tracking in GitHub/Weights & Biases to support auditability and reproducibility.
6) A dashboard summarizing key metrics and visualizations (e.g., anomaly scores, alert counts, alarm burden trends, and model comparison summaries).
7) Delieverables will include documented methology, reproducible code and environment, and final results.

Projected Project End Date: 1/24/2026

Sources: Dataset: Publically available Pump Sensor Data (pump_sensor_data) by UnknownClass from Kaggle.

Supporiting Sources: Documentation and Metadata provided on the Kaggle dataset page; Official documentation for tools, systems, and platforms used in the project (e.g. Python Libraries, Docker, Kafka, PostgreSQL, etc).

Additional references identified during the development and analysis will be cited in the final project report/results writeup.

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Human Subjects or Proprietary Information
Does your project involve the potential use of human subjects? (Y/N): No
Does your project involve the potential use of proprietary company information? (Y/N): No

STUDENT SIGNATURE

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By signing and submitting this form, you acknowledge that any cost associated with the development and execution of your data analytics solution will be your (the student) responsibility.

TO BE COMPLETED BY AN INSTRUCTOR

The capstone topic is approved by an instructor.

**COURSE INSTRUCTOR SIGNATURE:**

Jim Ashe, Ph.D. Mathematics

**COURSE INSTRUCTOR APPROVAL DATE:**

Sunday, December 28, 2025